

cudaampicommon.c

- float [computeDevPerformance](#) (double period_us)
Compute device performance based on the period between two events.
- float [getGPUpower](#) (int gpuid)
Get the power usage of a GPU.
- cudaError_t [__cudampi__getCpuFreeThreads](#) (int *count)
Get the number of free CPU threads.
- cudaError_t [getCpuEnergyUsed](#) (float *lastEnergyMeasured, float *energyUsed)
Get the energy used by the CPU.
- void [initializeCpuEnergyMeasurement](#) (int *isInitialCpuEnergyMeasured, omp_lock_t *cpuEnergyLock, float *cpuLastEnergyMeasured)
Initialize CPU energy measurement.

cudaampilib.c

- [TAILQ_HEAD](#) (memcpy_queue_head, memcpy_queue_entry)
- unsigned long [__cudampi__getCurrentBatchSize](#) ()
Gets the current batch size.
- void [__cudampi__recordBatchSizeForDeviceStats](#) (unsigned long batchsize)
Records the batch size for device statistics.
- void [__cudampi__finishDeviceStatsMeasurement](#) (double elapsedTimeSeconds)
Finishes device statistics measurement.
- int [__cudampi__readyForCpuBatchSizeScaling](#) ()
Checks if the system is ready for CPU batch size scaling.
- void [__cudampi__scaleCpuBatchSize](#) ()
Scales the CPU batch size.
- int [getMsgCounter](#) ()
Gets a unique message counter for asynchronous messages.
- void [initiateAsyncRecv](#) (void *dst, unsigned long count, int counter)
Initiates an asynchronous receive operation.
- void [initiateAsyncSendCpu](#) (const void *src, unsigned long count, int counter)
Initiates an asynchronous send operation for the CPU.
- void [waitForAsyncSendResponse](#) (int counter)
Waits for an asynchronous send response.
- void [process_queue](#) ()
Processes the memcpy queue.
- void [__cudampi__setglobalpowerlimit](#) (float powerlimit)
Sets the global power limit.
- float [__cudampi__gettotalpowerofselecteddevices](#) ()
Gets the total power of selected devices.
- int [__cudampi__selectdevicesforpowerlimit_greedy](#) ()
Selects devices for the power limit using a greedy strategy.
- __cudampi__batch_pointer [__cudampi__getnextchunkindex_enableddevices](#) (long long *globalcounter, unsigned long batchsize, long long max)
Gets the next chunk index for enabled devices.
- __cudampi__batch_pointer [__cudampi__getnextchunkindex_alldevices](#) (long long *globalcounter, unsigned long batchsize, long long max)
Gets the next chunk index for all devices.
- int [__cudampi__isdeviceenabled](#) (int deviceid)
Checks if a device is enabled.

- `cudaError_t __cudamp__getDeviceCount (int *count)`
Gets the total device count.
- `cudaError_t __cudamp__cudaGetDeviceCount (int *count)`
Gets the GPU device count.
- `cudaError_t __cudamp__cpuGetDeviceCount (int *count)`
Gets the CPU device count.
- `void __cudamp__initializeMPI (int argc, char **argv)`
Initializes MPI.
- `void __cudamp__terminateMPI ()`
Terminates MPI.
- `int __cudamp__gettargetGPU (int device)`
Gets the target GPU ID for a device.
- `int __cudamp__gettargetMPIrank (int device)`
Gets the target MPI rank for a device.
- `cudaError_t __cudamp__cudaMalloc (void **devPtr, size_t size)`
Allocates memory on the GPU.
- `cudaError_t __cudamp__cpuMalloc (void **devPtr, size_t size)`
Allocates memory on the CPU.
- `cudaError_t __cudamp__malloc (void **devPtr, size_t size)`
Allocates memory on the appropriate device (CPU or GPU).
- `cudaError_t __cudamp__cudaFree (void *devPtr)`
Frees memory on the GPU.
- `cudaError_t __cudamp__cpuFree (void *devPtr)`
Frees memory on the CPU.
- `cudaError_t __cudamp__free (void *devPtr)`
Frees memory on the appropriate device (CPU or GPU).
- `cudaError_t __cudamp__cudaDeviceSynchronize (void)`
Synchronizes the device (GPU or CPU).
- `cudaError_t __cudamp__deviceSynchronize (void)`
Synchronizes the device (GPU or CPU).
- `cudaError_t __cudamp__cudaSetDevice (int device)`
Sets the device (GPU or CPU).
- `int __cudamp__isCpu ()`
Checks if the current device is a CPU.
- `cudaError_t __cudamp__setDevice (int device)`
Sets the device (CPU or GPU).
- `cudaError_t __cudamp__cudaMemcpy (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind)`
Copies memory between host and device (GPU).
- `cudaError_t __cudamp__cpuMemcpy (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind)`
Copies memory between host and device (CPU).
- `cudaError_t __cudamp__cudaMemcpyAsync (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind, cudaStream_t stream)`
Asynchronously copies memory between host and device (GPU).
- `cudaError_t __cudamp__cpuMemcpyAsync (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind, cudaStream_t stream)`
Asynchronously copies memory between host and device (CPU).
- `void __cudamp__cudaKernelInStream (void *devPtr, unsigned long batchsize, cudaStream_t stream)`
Launches a CUDA kernel in a specific stream.
- `void __cudamp__cudaKernel (void *devPtr, unsigned long batchsize)`
Launches a CUDA kernel.

- void [__cudamp__cpuKernelInStream](#) (void *devPtr, unsigned long batchSize, cudaStream_t stream)
Launches a CPU kernel in a specific stream.
- void [__cudamp__cpuKernel](#) (void *devPtr, unsigned long batchSize)
Launches a CPU kernel.
- cudaError_t [__cudamp__cudaStreamCreate](#) (cudaStream_t *pStream)
Creates a CUDA stream.
- cudaError_t [__cudamp__cudaStreamDestroy](#) (cudaStream_t stream)
Destroys a CUDA stream.
- cudaError_t [__cudamp__cpuStreamCreate](#) (cudaStream_t *pStream)
Creates a CPU stream.
- cudaError_t [__cudamp__cpuStreamDestroy](#) (cudaStream_t stream)
Destroys a CPU stream.
- cudaError_t [__cudamp__streamCreate](#) (cudaStream_t *stream)
Creates a stream (CPU or GPU).
- cudaError_t [__cudamp__streamDestroy](#) (cudaStream_t stream)
Destroys a stream (CPU or GPU).
- cudaError_t [__cudamp__memcpyAsync](#) (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind, cudaStream_t stream)
Asynchronously copies memory between host and device (CPU or GPU).
- cudaError_t [__cudamp__memcpy](#) (void *dst, const void *src, size_t count, enum cudaMemcpyKind kind)
Copies memory between host and device (CPU or GPU).
- void [__cudamp__kernelInStream](#) (void *devPtr, cudaStream_t stream)
Launches a kernel in a specific stream (CPU or GPU).
- void [__cudamp__kernel](#) (void *devPtr)
Launches a kernel (CPU or GPU).

cudampislave.c

- [TAILQ_HEAD](#) (task_queue_head, task_queue_entry)
- void [launchkernel](#) (void *devPtr, unsigned long batchSize)
- void [launchkernelinstream](#) (void *devPtr, unsigned long batchSize, cudaStream_t stream)
- void [launchcpukernel](#) (void *devPtr, unsigned long batchSize, int thread_count)
- allocatedGpuMemcpyBuffer [allocateGpuMemcpyBuffer](#) (globalGpuMemcpyBuffer *global, unsigned long long count)
Allocates a GPU memory copy buffer.
- void [freeGpuMemcpyBuffer](#) (allocatedGpuMemcpyBuffer *allocatedBuffer)
Frees a GPU memory copy buffer.
- void [updateGlobalGpuMemcpyBuffer](#) (globalGpuMemcpyBuffer *global)
Updates the global GPU memory copy buffer.
- void [initializeGlobalGpuMemcpyBuffer](#) (globalGpuMemcpyBuffer *global, unsigned long count)
Initializes the global GPU memory copy buffer.
- void [freeGlobalGpuMemcpyBuffer](#) (globalGpuMemcpyBuffer *global)
Frees the global GPU memory copy buffer.
- void [cpuSynchronize](#) ()
Synchronizes CPU tasks.
- void [cpuHostToDeviceTaskAsync](#) (void *arg)
Asynchronously transfers data from host to device on the CPU.
- void [cpuDeviceToHostTaskAsync](#) (void *arg)
Asynchronously transfers data from device to host on the CPU.
- void [gpuHostToDeviceTaskAsync](#) (void *arg)

- Asynchronously transfers data from host to device on the GPU.*
- void `gpuDeviceToHostTaskAsync` (void *arg)
 - Asynchronously transfers data from device to host on the GPU.*
- void `logGpuMemcpyError` (cudaError_t e, int tag)
 - Logs GPU memory copy errors.*
- void `cpuLaunchKernelTask` (void *arg)
 - Launches a CPU kernel task.*
- void `allocateCpuTaskInStream` (void(*task_func)(void *), void *arg, unsigned long stream)
 - Allocates a CPU task in a specific stream.*
- void `scheduleCpuTask` (task_queue_entry_t *current_task_queue_entry, unsigned long stream)
 - Schedules a CPU task.*
- void `cpuTaskLauncher` (unsigned long stream)
 - Launches CPU tasks in a specific stream.*
- int `main` (int argc, char **argv)
 - Main function for the slave process.*